

## Calc 1 Lecture 2: Exponentials, logarithms and inverses

An exponential function is a function of the form

$$f(x) = a^x$$

where  $a$  is a positive constant.

Note: For exponential functions  $f(x) = a^x$ , we require  $a$  to be positive.  
Why is that?

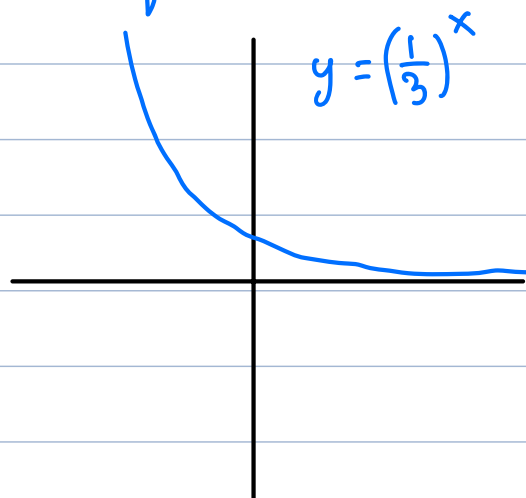
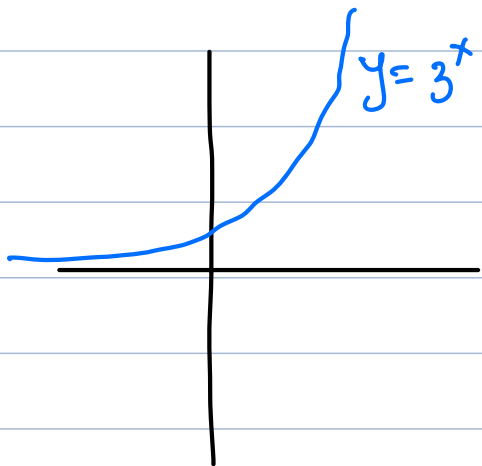
Example:  $(-1)^{1/2} = \sqrt{-1}$  undefined

Some observations:

a)  $1^x = 1$  for all  $x$

b)  $(\frac{1}{a})^x = (a^{-1})^x = a^{-x}$

Therefore, for example, the graph of  $y = (\frac{1}{3})^x$  is a reflection about the  $y$ -axis of the graph of  $y = 3^x$



Trends of growth : • Suppose  $a > 1$ . Then the function

$y = a^x$  exhibits exponential growth. This means that, in the long term, it grows faster than any polynomial.

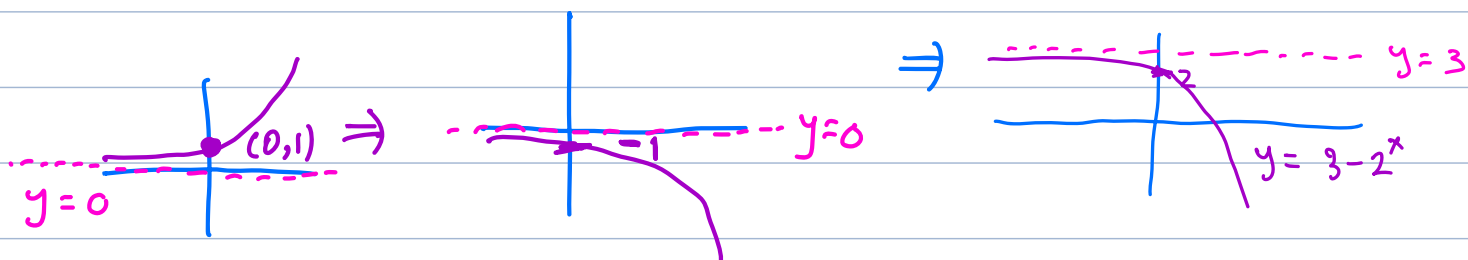
in contrast if  $a < 1$ , it exhibits exponential decay.

Example sketch the graph of the function  $y = 3 - 2^x$  and determine its domain and range.

$y = 2^x \Rightarrow$   
vertical reflection

$y = -2^x \Rightarrow$   
shift up

$$y = 3 - 2^x$$



domain:  $(-\infty, \infty)$   
range:  $(-\infty, 3)$

## Laws of exponents:

- $a^{x+y} = a^x \cdot a^y$
- $a^{x-y} = \frac{a^x}{a^y}$
- $(a^x)^y = a^{x \cdot y}$
- $(ab)^x = a^x \cdot b^x$

Example: Use laws of exponents to simplify

$$(3x)^4 \cdot (\sqrt{x})^{-2}$$

Solution:

$$(3x)^4 = 3^4 \cdot x^4 = 81x^4$$

$$(\sqrt{x})^{-2} = (x^{1/2})^{-2} = x^{\frac{1}{2} \cdot (-2)} = x^{-1} = \frac{1}{x}$$

$$81x^4 \cdot \frac{1}{x} = \boxed{81x^3}$$

## Application of exponentials: population growth.

Example: A colony of bacteria starts with 1000 cells and doubles in size every hour. Write an expression for population size at time  $t$ .

↖ in hours

$$P(t) = 1000 \cdot 2^t$$

$$\text{check: } P(0) = 1000 \cdot 2^0 = 1000 \quad \checkmark$$

$$\text{check: } P(t+1) = 1000 \cdot 2^{t+1} = 1000 \cdot 2 \cdot 2^t = 2P(t) \quad \checkmark$$

## Application of exponentials: radioactive decay.

You start with 100g of a material undergoing decay.

Suppose the "half-life" is 2 days. } halves in size every two days  
Write a function representing the mass of material after  $t$  days.

Idea:

$$m(t) = 100 \cdot \left(\frac{1}{2}\right)^{\frac{t}{2}}$$

$$100 \cdot \left(\frac{1}{2}\right)^{t/2}$$

$$\text{check: } m(0) = 100$$

$$m(t+2) = \frac{1}{2} m(t)$$

e: A special base.

$$(e = 2.71\dots)$$

Exponentials are often expressed in base e.

What is special about e?

Short answer:  $f(x) = e^x$  is an exponential function  
satisfying  $f'(x) = f(x)$

(more on this when we  
cover derivatives)

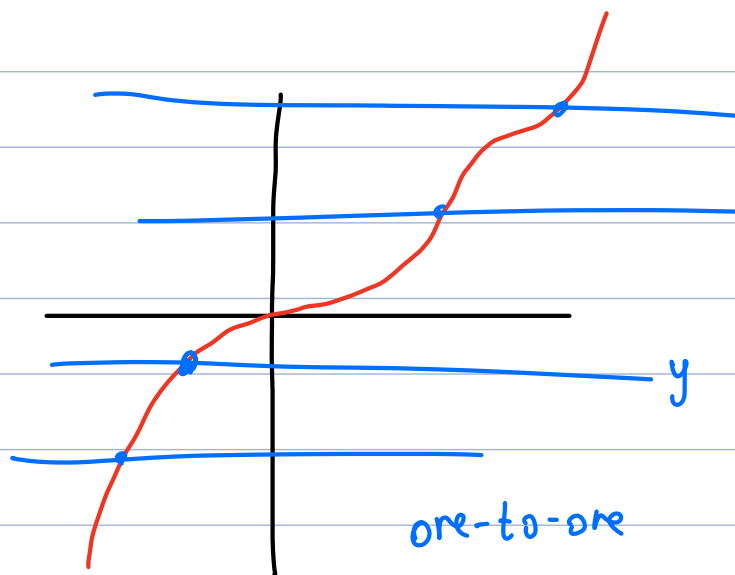
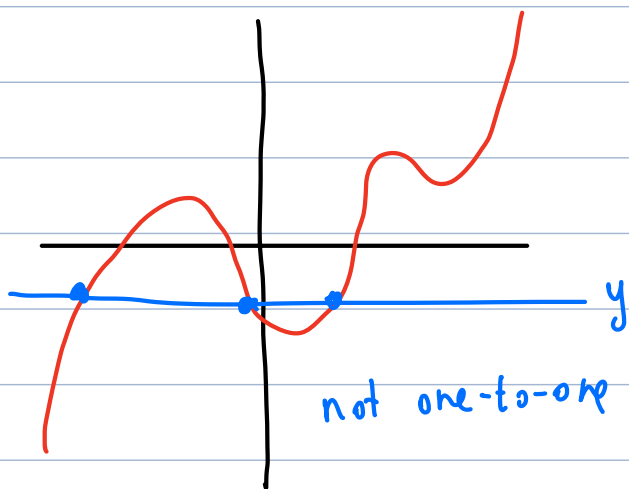
# Inverse functions

Next we consider inverses of functions.

In order for a function to have an inverse, it needs to be one-to-one.

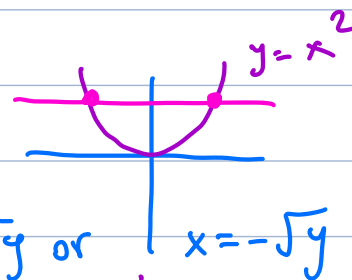
Definition: A function  $f(x)$  is one-to-one if it passes the horizontal line test: Every horizontal line intersects the graph of  $f$  at most once.

In other words, for each possible value of  $y$ , there is at most one value of  $x$  for which  $y = f(x)$ .



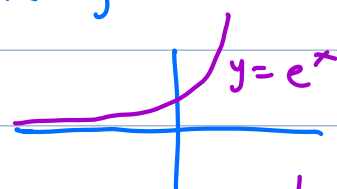
Example: a) Is  $y = x^2$  one-to-one?

No



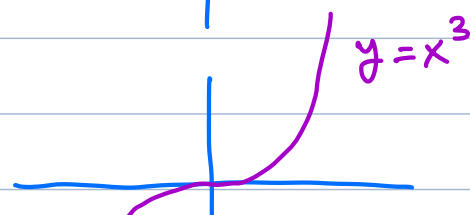
b) Is  $y = e^x$  one-to-one?

Yes



c) Is  $y = x^3$  one-to-one?

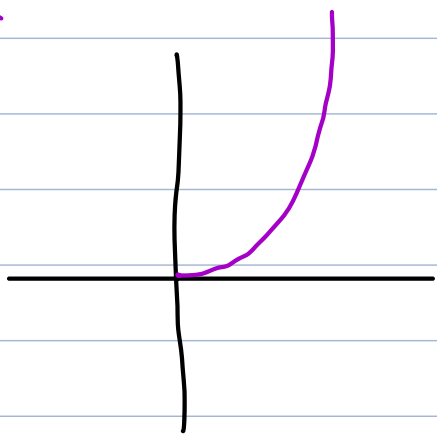
Yes



$$y = x^3 \Rightarrow x = y^{1/3}$$

Note: Sometimes, even if a function is not one-to-one, we can make it one-to-one by restricting its domain.

For example,  $y = x^2$  becomes one-to-one if we restrict it to only non-negative numbers



Inverse functions. If  $f(x)$  is a one-to-one function, then its inverse function is a function  $f^{-1}(x)$  satisfying

$$y = f(x) \iff x = f^{-1}(y)$$

" $f^{-1}$  undoes whatever  $f$  does"

Caution:  $f^{-1}(x)$  DOES NOT mean  $\frac{1}{f(x)}$ .

In other words: to find  $f^{-1}$ , we "switch the roles of  $x$  and  $y$ " in the definition of  $f$ .

Example:  $y = f(x) = x^3$

$y = x^3 \Rightarrow$  inverse function  $x = y^3 \Rightarrow y = x^{1/3}$

$$f^{-1}(x) = x^{1/3}$$

Cancellation equations:

i)  $f^{-1}(f(x)) = x$

ii)  $f(f^{-1}(x)) = x$

Domain and range: Taking the inverse switches domain and range:

$$\text{domain}(f^{-1}) = \text{range}(f)$$

$$\text{range}(f^{-1}) = \text{domain}(f)$$



How to find the inverse function of  $f$ :

- step 0: make sure  $f$  is one-to-one. otherwise there is no inverse.
- Step 1: write  $y = f(x)$
- step 2: solve this equation for  $x$  as a function of  $y$  (if possible)
- Step 3: interchange  $x$  and  $y$ . This will give you a formula for  $y = f^{-1}(x)$
- step 4 (optional): Check your work by verifying the cancellation identities

Example Find the inverse function:

a)  $y = 3x + 2 = f(x)$

b)  $y = x^3 - 10$

solutions

a) step 1:  $y = 3x + 2$

step 2: solve for  $x$ .  $y - 2 = 3x \Rightarrow \frac{y-2}{3} = x$

step 3:  $y = f^{-1}(x) = \frac{x-2}{3}$

step 4:  $f(f^{-1}(x)) = 3\left(\frac{x-2}{3}\right) + 2 = x - 2 + 2 = x. \checkmark$

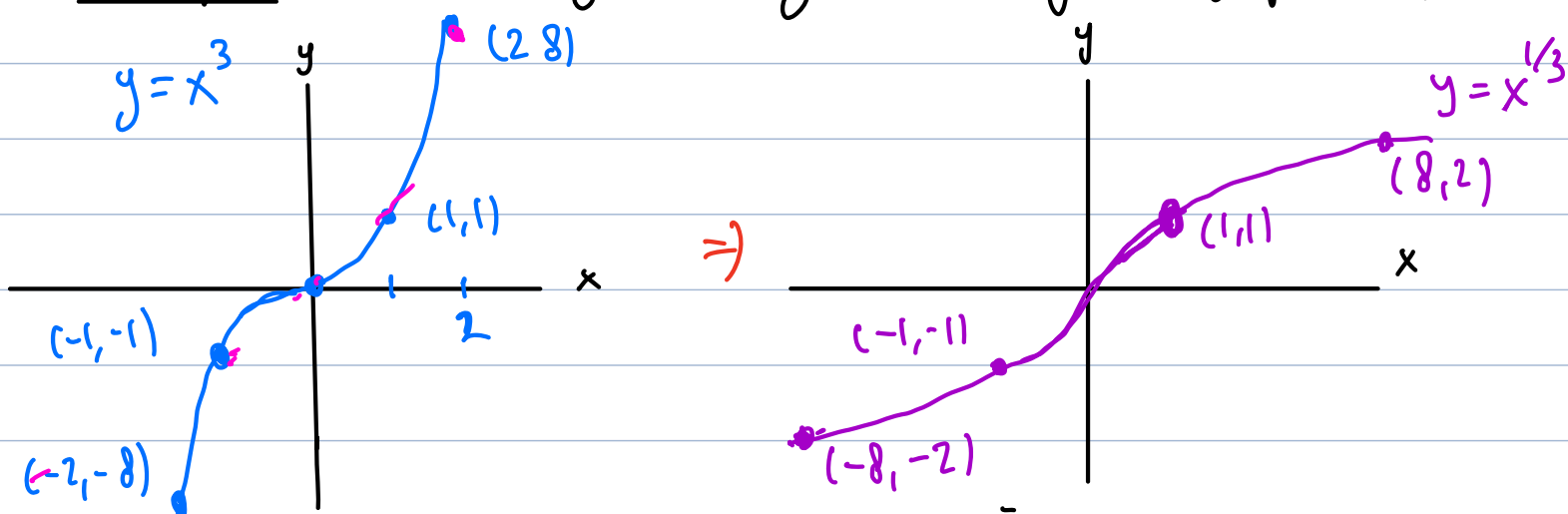
Note: you can also get the answer more intuitively

"the inverse undoes everything the original function does"

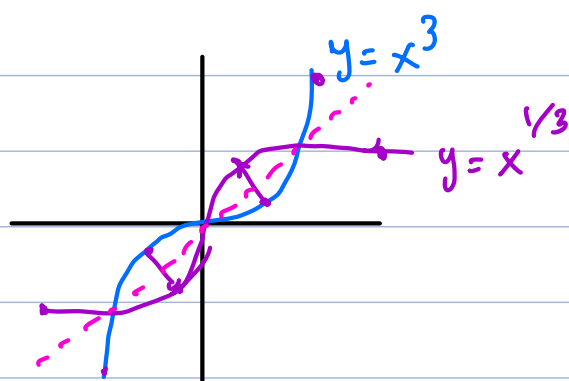
Graphing inverses: Even if you cannot determine a formula for the inverse of  $f$ , you can sketch its graph.

Idea: if  $(x, y)$  is on the graph of  $f$ , then  $(y, x)$  is on the graph of  $f^{-1}$ .

Example: sketch  $y = x^{1/3}$  by consulting the graph of  $y = x^3$ .



observation: In fact, the graph of  $f^{-1}(x)$  is a reflection of the graph of  $f(x)$  about the line  $y = x$ .

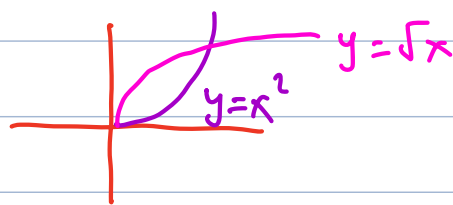
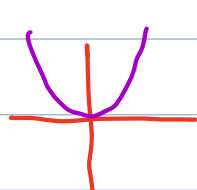


Question : a) does  $y = x^2$  have an inverse?

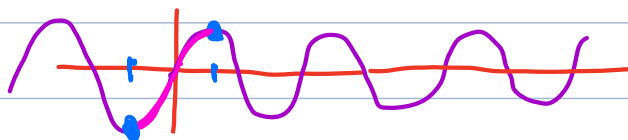
b) does  $y = \sin(x)$  have an inverse?

Answers:

a) NO, unless we restrict the domain to non-negatives



b)



not one-to-one NO, unless  
restrict to  $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

$\sin^{-1}(x)$  is the inverse of  $\sin(x)$   $\left\{ -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \right\}$   
range of  $\sin^{-1}(x)$   
domain of  $\sin^{-1}$  is  $[-1, 1]$

Conventions for inverse trig functions

•  $\sin^{-1}(x)$  has range  $\left[ -\frac{\pi}{2}, \frac{\pi}{2} \right]$

•  $\cos^{-1}(x)$  has range  $[0, \pi]$

•  $\cot^{-1}(x)$  has range  $[0, \pi]$

•  $\tan^{-1}(x)$  has range  $\left[ -\frac{\pi}{2}, \frac{\pi}{2} \right]$

•  $\sec^{-1}(x)$  has range  $[0, \pi]$

•  $\csc^{-1}(x)$  has range  $\left[ -\frac{\pi}{2}, \frac{\pi}{2} \right]$

# Logarithms

The inverse of an exponential function is a logarithmic function

By definition,  $\log_b x$  denotes the inverse of  $b^x$

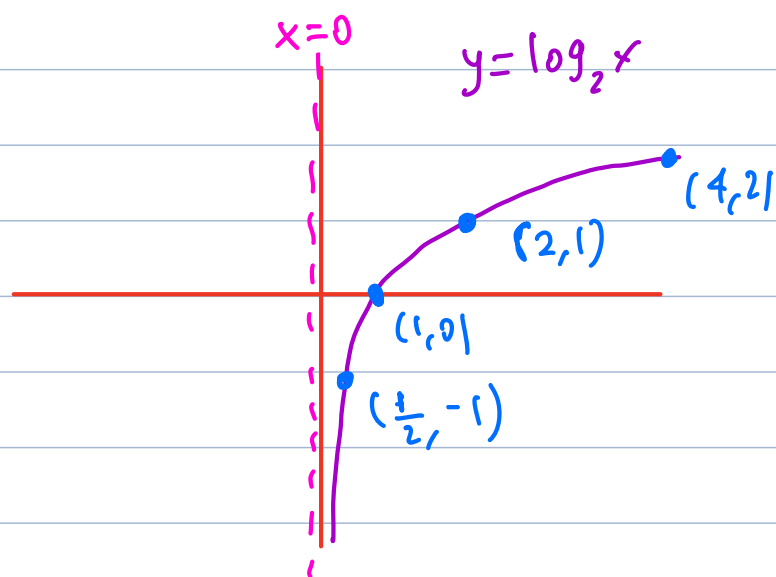
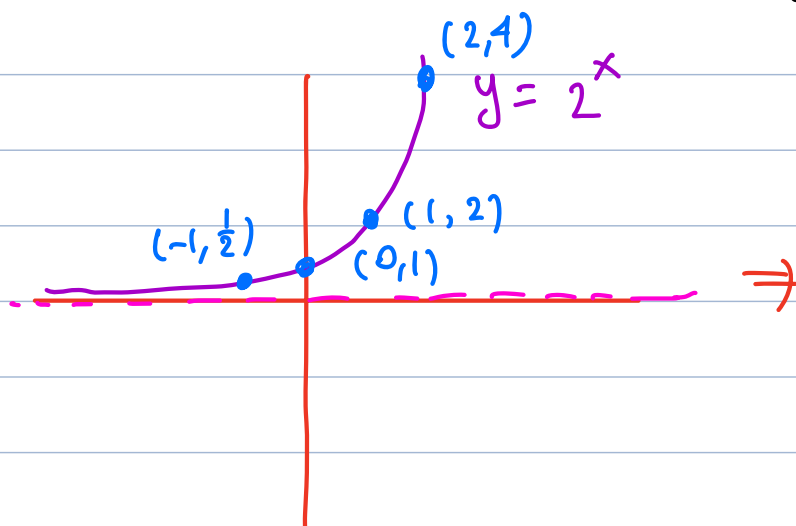
In words: " $\log_b x$  is the exponent necessary to raise  $b$  to in order to get  $x$ "

e.g.  $\log_2 8 = 3$  because  $2^3 = 8$

By the cancellation identities, this means:

- $b^{\log_b x} = x$
- $\log_b (b^x) = x$

Question: sketch  $y = \log_2 x$  by referring to the graph of  $y = 2^x$



Note:  $\log_b x$  is only defined if  $x > 0$

## Log properties

$$1. \log_b(xy) = \log_b x + \log_b y$$

$$2. \log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$$

$$3. \log_b(x^r) = r \log_b x$$

Example:  $\underbrace{\log_2 8}_3 + \underbrace{\log_2 4}_2 = 5 = \log_2(8 \cdot 4) = \log_2(32) \quad \checkmark$

## Deriving log properties

Can be derived using exponentiation properties

## Example Back to population growth.

We saw in previous example, we can model

$$P(t) = 1000 \cdot 2^t \quad (t = \text{hours})$$

Question: after how many hours does the population reach 1,000,000?

$$P(t) = 1000000 \Rightarrow 1000 \cdot 2^t = 1000000 \Rightarrow 2^t = 1000$$

d.v.d  
by 1000

take  $\log_2$

$\Rightarrow$

$$\underbrace{\log_2(2^t)}_t = \log_2(1000)$$

$\Rightarrow$

$$\boxed{t = \log_2 1000}$$

## Natural log:

Just as  $e$  is commonly used as base for exponentiation it's also often used as base for log:

$$\ln(x) = \log_e(x)$$

Example Solve the equation  $e^{5-3x} = 10$

$$\ln(e^{5-3x}) = \ln(10)$$

$$5-3x = \ln(10) \Rightarrow 3x = 5 - \ln(10)$$

$$x = \frac{5 - \ln(10)}{3}$$

Example Express  $\ln(5) + \frac{1}{2}\ln(7)$  as a single logarithm.

$$\ln(5) + \ln(7^{1/2}) = \ln(5 \cdot 7^{1/2}) = \boxed{\ln(5\sqrt{7})}$$

change of base If we want to compare two logarithms, we should be able to convert from one base to another. This is done by:

$$\log_a x = \frac{\log_b x}{\log_b a}$$

specifically, if we choose  $b = e$ ,

$$\log_a x = \frac{\ln x}{\ln a}$$

Example a) Rewrite  $\log_2(x-3)$  in terms of  $\ln$

$$= \boxed{\frac{\ln(x-3)}{\ln(2)}}$$

b) Can you rewrite  $2^x$  as an exponential with base  $e$ ?

$$2^x = e^?$$

$$2 = e^{\ln 2} \Rightarrow$$

raise to the  $x$

$$2^x = (e^{\ln 2})^x = \boxed{e^{(\ln 2 \cdot x)}}$$

## Working with inverse trig functions

Example: Find an expression for  $\sin(\cos^{-1}(x))$

Idea:  $\cos^{-1}(x) = \theta$  "the angle whose cosine is  $x$ "

$$\cos^{-1}(x) = \theta \iff \boxed{x = \cos \theta}$$

represent this in  
a triangle

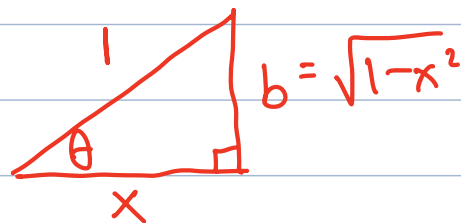
$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = x = \frac{x}{1}$$

$$\underline{\sin(\cos^{-1}(x)) = \sin \theta}$$

$$= \frac{\text{opposite}}{\text{hypotenuse}} = \frac{\sqrt{1-x^2}}{1}$$

$$= \boxed{\sqrt{1-x^2}}$$

In conclusion:  $\boxed{\sin(\cos^{-1}(x)) = \sqrt{1-x^2}}$



Pythagoras:  $x^2 + b^2 = 1$

$$b^2 = 1 - x^2$$

$$b = \sqrt{1-x^2}$$